

COMP2300 Set 5 Practice Exam

Topics: Multi-Cycle Microarchitecture, FSM Control, Performance,
Microprogramming, Pipelining Intro

Time: 120 minutes

Total: 100 marks

Instructions

- Answer in English.
- Part A is multiple-choice (concept-focused).
- Part B is computational. Show key steps.
- Part C contains past exam questions with sources.
- Assume 32-bit ARM unless stated otherwise.

Part A: Multiple Choice (30 marks, 3 marks each)

Choose exactly one option for each question.

A1. The main motivation for moving from a single-cycle CPU to a multi-cycle CPU is to:

- (A) Guarantee $CPI = 1$ for every instruction
- (B) Shorten the clock period by breaking execution into states
- (C) Remove the need for a control unit
- (D) Make instruction encoding simpler

A2. From the ISA point of view, the intermediate states used by a multi-cycle implementation are:

- (A) Architecturally visible to the assembly programmer
- (B) Required to be stored in general-purpose registers
- (C) Programmer-invisible microarchitectural state
- (D) Part of the instruction opcode

A3. Which of the following is a downside of a multi-cycle design?

- (A) Hardware resources cannot be reused
- (B) Additional state registers and sequencing overhead are needed
- (C) Branches cannot be implemented
- (D) CPI is fixed at 1

A4. In the final multi-cycle ARM controller discussed in Set 5, which instruction class takes 5 cycles?

- (A) B
 - (B) Data-processing
 - (C) STR
 - (D) LDR
- A5.** When R15 is read as a source operand, the register file should provide:
- (A) The current PC value
 - (B) $PC + 4$
 - (C) $PC + 8$
 - (D) The branch target address
- A6.** For an ARM B instruction, ExtImm is formed by:
- (A) Zero-extending imm12
 - (B) Sign-extending imm24 after shifting left by 2
 - (C) Sign-extending imm8
 - (D) Adding 8 directly to the PC
- A7.** In microprogrammed control, the control signals for the current state are stored in a:
- (A) Branch predictor
 - (B) Microinstruction
 - (C) Register file entry
 - (D) Cache line
- A8.** The component that determines the address of the next microinstruction is the:
- (A) Microsequencer
 - (B) ALU
 - (C) Data memory
 - (D) Extend block
- A9.** The main limitation of a multi-cycle processor that motivates pipelining is:
- (A) Instructions become variable-width
 - (B) Hardware resources are often idle, limiting concurrency
 - (C) Immediate values become too small
 - (D) PC can no longer be updated on branches
- A10.** Pipelining primarily improves:
- (A) The ISA semantics
 - (B) Throughput by overlapping different stages of different instructions
 - (C) The width of immediate fields
 - (D) The number of architectural registers

Part B: Computational Problems (50 marks)

B1. State sequences and cycle counts (8 marks)

In the final multi-cycle FSM, the available states include `Fetch`, `Decode`, `MemAdr`, `MemRead`, `MemWB`, `MemWrite`, `ExecuteI`, `ExecuteR`, `ALUWB`, and `Branch`.

- (a) Write the state sequence and cycle count for each of the following instruction classes: `LDR`, `STR`, data-processing immediate, and `B`.
- (b) How many cycles are needed to execute the following 6-instruction sequence on this processor? What is the average CPI?

```
LDR R0, [R1, #0]
ADD R2, R2, #1
STR R2, [R1, #4]
B label
ADD R3, R3, R4
LDR R5, [R6, #8]
```

B2. PC, R15, and branch target calculation (8 marks)

An instruction is located at address `0x80A0`.

- (a) If the instruction is `ADD R1, R15, R2`, what value should be supplied when `R15` is read in the register-read step?
- (b) If the instruction is `B` with `imm24 = 0x000003`, compute `ExtImm` and the branch target address.

B3. Decoder outputs (8 marks)

The decoder uses the following relations:

$$\text{RegSrc0} = (\text{Op} == 10_2), \quad \text{RegSrc1} = (\text{Op} == 01_2), \quad \text{ImmSrc} = \text{Op}$$

Fill in the outputs for each instruction class below:

Instruction class	Op	RegSrc1:0	ImmSrc
LDR	01	?	?
STR	01	?	?
DP immediate	00	?	?
DP register	00	?	?
B	10	?	?

B4. Average CPI and total cycles (8 marks)

A workload contains 25% loads, 10% stores, 13% branches, and 52% data-processing instructions. Assume:

$$CPI_{LDR} = 5, \quad CPI_{STR} = 4, \quad CPI_B = 3, \quad CPI_{DP} = 4$$

- (a) Compute the average CPI.
- (b) If the program has 50 million instructions, how many total cycles does it take?

B5. Critical path and clock frequency (8 marks)

Assume the multi-cycle critical-path formula is

$$T_{c2} = t_{pcq} + 2t_{mux} + \max(t_{ALU} + t_{mux}, t_{mem}) + t_{setup}$$

and the delays are:

$$t_{pcq} = 40 \text{ ps}, \quad t_{mux} = 25 \text{ ps}, \quad t_{ALU} = 120 \text{ ps}, \quad t_{mem} = 200 \text{ ps}, \quad t_{setup} = 50 \text{ ps}$$

- (a) Compute T_{c2} .
- (b) Compute the corresponding maximum clock frequency in GHz.

B6. Execution time (10 marks)

A program executes 100×10^9 instructions on a multi-cycle ARM processor with:

$$CPI = 4.12, \quad T_c = 340 \text{ ps}$$

- (a) Compute the execution time in seconds.
- (b) Suppose a revised design keeps the same CPI but reduces the clock period to 300 ps. What is the new execution time?
- (c) What is the speedup of the revised design over the original design?

Part C: Past Exam Questions (Source-Tagged) (20 marks)**C1. ISA vs Microarchitecture (Source: exam24px1.pdf, Q1f–Q1i, adapted) (5 marks)**

For each item below, classify it as **ISA** or **Microarchitecture**.

- (a) Whether the ALU uses a ripple-carry adder or a carry-save adder
- (b) Instruction width
- (c) Whether the CPU uses branch prediction
- (d) Whether the control unit is hardwired or microprogrammed

C2. Multi-Cycle Execution Time (Source: exam2025-main.pdf, Q12) (5 marks)

A program consists of 2 billion instructions, of which 50% are ALU instructions, 25% are memory instructions, and 25% are branch instructions. On a 1 GHz multi-cycle CPU, an ALU instruction takes 6 cycles, each memory instruction takes 8 cycles, and each branch instruction takes 4 cycles. What is the execution time in seconds?

C3. Single-Cycle vs Pipelining (Source: exam24px2.pdf, Q6, adapted) (5 marks)

State two drawbacks of a single-cycle processor microarchitecture. Explain how instruction pipelining addresses these drawbacks. Also define instruction-level parallelism (ILP), and name two types of microarchitectures that exploit ILP.

C4. ILP Ranking (Source: 2023_Final_Exam.pdf, Bonus Q1, adapted) (5 marks)

Rank the following microarchitectures from **highest** to **lowest** potential to exploit instruction-level parallelism. Briefly justify your ranking.

- (A) Scalar in-order pipeline with forwarding and hazard detection
- (B) Scalar in-order pipeline without forwarding or hazard detection (compiler inserts NOPs)
- (C) Multi-cycle processor